* principle
Chromatography is based on the principle of differential partitioning, where the component of a mixture are separated based on their interactions with a stationary phase & a mobile phase.

* Types of Chromatography
1. Paper chromatography: uses paper as the stationary phase and a solvent as the mobile phase.
2. Thin layer chromatography (TLC): Uses a thin layer of silica gel or alumina as the stationary phase & solvent as the mobile phase.
3. Column chromatography: uses a column packed with a stationary phase & a solvent as the mobile phase.
4. Gas chromatography (GC): uses a gas as the mobile phase & a stationary phase coated on a column.
5. Liquid chromatography (LC): uses a liquid as the mobile phase & a stationary phase coated on a column.

Applications
Chromatography has a wide range of applications in various fields, including:
1. Pharmaceuticals: To separate & identify active ingredients & impurities.
2. Forensic Science: To analyze evidence & identify substances.
3. Environmental monitoring: To detect & quantify pollutants in air, water & soil
4. Food & Beverages: To analyze the composition of foods & beverages.
5. Biotechnology: To separate & purify biomolecules, such as proteins & nucleic acids

Advantages:
1. High Sensitivity & Selectivity: chromatography can detect & separate very small amounts of substances.
2. Fast analysis: Chromatography can analyze complex mixtures quickly & efficiently
2. High accuracy: chromatography can provide accurate and reliable results.